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RESEARCH ARTICLE

EEMLCR: Energy-Efficient Machine Learning-Based Clustering and Routing for Wireless Sensor Networks

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ABSTRACT Wireless Sensor Networks (WSNs) offer a powerful technology for sensing and transmitting data across vast geographical regions. However, limitations inherent to WSNs, such as low-power sensor units, communication constraints, and limited processing capabilities, can significantly impact their lifespan. To address these limitations and enhance the energy efficiency of WSNs, it is often necessary to divide sensors into clusters and establish routing to conserve energy. Machine learning algorithms can potentially automate these processes, minimizing energy consumption and extending network lifetime. This research investigates the application of machine learning algorithms, specifically Q-learning and K-means clustering, to propose the Energy-Efficient Machine Learning-based Clustering and Routing (EEMLCR) method for WSNs. This method facilitates cluster formation and routing path selection. The proposed method is compared with the well-established LEACH algorithm and two multi-hop variants, DMHT LEACH and EDMHT LEACH to validate its effectiveness. Our experimental results demonstrate the effectiveness of EEMLCR compared to LEACH and its multi-hop variants (DMHT LEACH and EDMHT LEACH). After 600 rounds in networks comprising 400 nodes, EEMLCR showed significant improvements in key performance metrics. These include increased alive nodes, reduced average energy consumption, higher remaining energy levels, and improved packet reception. Additionally, we compared EEMLCR with recent state-of-the-art algorithms such as EECDA and CMML, where our method demonstrated comparable or superior performance in terms of network lifetime and energy efficiency. By optimizing clustering and routing strategies, WSNs can reduce energy consumption, leading to more efficient utilization of the limited energy resources available to sensor nodes. The primary objective of this research is to contribute to the development of energy-efficient WSNs by leveraging machine learning algorithms for data routing and the cluster-based organization of sensor nodes.

INDEX TERMS Clustering algorithms, energy efficiency, machine learning, multi-hop communication, routing protocols, wireless sensor networks.

I. INTRODUCTION

Wireless Sensor Networks (WSNs) are self-organizing, self-healing, and self-configurable networks comprising

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homogeneous or heterogeneous sensor nodes. These nodes can be mobile or static and are deployed in an unattended geographical area where human intervention may be impossible or difficult [1]. WSNs sense environmental factors such as temperature, humidity, pressure, vibration, and lighting in various locations, including forests, near active volcanoes, and cold regions. There is no fixed network infrastructure in such areas, so WSNs possess self-organization, self-configuration, self-healing, and dynamic topology qualities [2].

WSN communication modes refer to how the data is transmitted from the source node to the destination node. Usually, the data is transmitted in two modes: single-hop or multi-hop. In single-hop WSN communication, there is a direct link between the source and destination device, so the data from one node to the other node travels only through a single networking device. The data travels between multiple networking devices in multi-hop transmission (see Figure 1). Because of the flexibility and scalability of multi-hop networks, most WSNs are deployed using multi-hop WSNs, which also preserve a good amount of energy and make efficient data routing [3], [4]. Table 1 compares the key features of single-hop and multi-hop WSNs. The range limitation in single-hop networks results in higher energy consumption because of long-distance communication, making them less energy-efficient than multi-hop networks. Multi-hop networks are highly scalable because of their ability to relay messages through multiple nodes, and they can cover larger areas and support more nodes than single-hop networks. Figure 1 shows the schematic for multi-hop communication in a cluster-based Wireless Sensor Network Configuration. In a cluster-based Wireless Sensor Network, multi-hop communication is facilitated by the mutual coordination among cluster heads of the adjoining clusters.

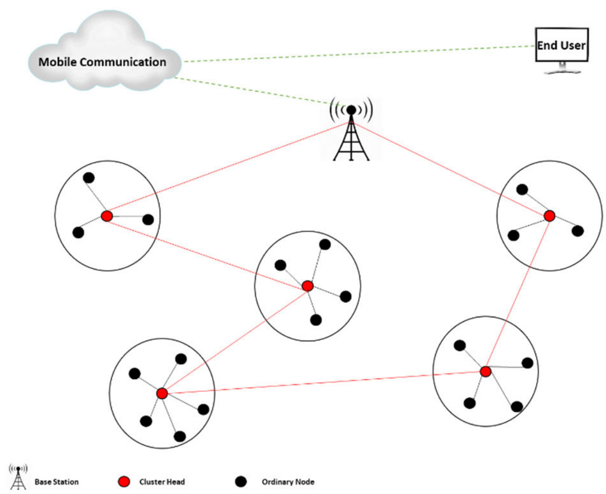

FIGURE 1. Typical multi-hop wireless sensor networks structure.

Another important difference between single-hop and multi-hop networks is seen in terms of network redundancy (Network redundancy is characterized by the superfluous replication of data conveyed by numerous sensors stemming from intersecting sensing domains, which consequently results in augmented energy expenditure and diminutions in operational efficacy within wireless sensor networks [5]). Multi-hop networks are less vulnerable to single-point failure as messages can be routed through multiple paths [6]. In contrast, single-hop networks are more vulnerable to single-point failures, which can cause significant issues in the network. Similarly, reliability is also a significant concern in WSNs. Multi-hop networks are more reliable than single-hop networks because of their use of redundant paths and their ability to compensate for node failures. Table 1 compares the single-hop and multi-hop communication configurations in WSNs.

TABLE 1. Comparison between key features of single-hop and multi-hop WSNs.

Features	SINGLE-HOP WSN	Multi-hop WSN
Range Limitation	Limited by transmission range of nodes	Not Limited by the transmission range of nodes, it can cover larger areas
Energy Consumption	High energy consumption due to long-distance	Lower energy consumption due to shorter distances, so nodes can be used for longer periods.
Scalability	Limited scalability due to range limitation	Highly scalable due to the ability to relay messages through multiple nodes
Network redundancy	Low network redundancy and failure of nodes can cause network failure	High network redundancy, messages can be routed through multiple paths, and failure of nodes can be compensated
Reliability	Lower reliability due to the potential for single-point	Higher reliability due to redundant paths, less vulnerable to single-point failure
Data latency	Lower data latency due to direct communication	Higher data latency due to multiple hops and the potential for congestion
Security	Lower security due to easy access to all nodes	Higher security due to multiple paths and the ability to encrypt messages

A typical WSN is composed of many sensing nodes and a sink. The sensor nodes monitor the environment in which they are deployed and gather the data. The collected data is then transmitted to the sink, which is connected to the server and transmits the data from nodes to the server [7]. Despite various advantages ranging from smart devices [8] to environmental surveillance, health monitoring, particularly in medicine [9], and easy deployment to monitor industries and other fields, these sensors also have numerous disadvantages. Because of sensor nodes' limited power, processing requirements, and high energy consumption, the lifetime and capabilities of WSN applications can be decreased [10]. Moreover, once these sensor nodes are deployed, the power that the nodes are equipped with is limited and unchangeable. This makes power and energy one of the most critical resources of a WSN, underscoring the importance of integrating power-efficient schemes to increase the lifetime and efficiency of the WSN. Thus, investigating how to efficiently utilize the limited resources, achieve load balancing among nodes, and extend the network lifetime is a critical issue in WSNs. This investigation becomes more relevant as power-efficient routing algorithms can greatly reduce energy consumption and extend the survival cycle of WSNs [11].

Conventional routing techniques aim to prolong the lifetime of WSNs by controlling data flow through specific node selection and clustering. The high hierarchical cluster head nodes are used by the low-energy adaptive clustering hierarchy (LEACH) routing algorithm to evenly receive and process the sensing data inside each cluster once the sensor nodes have been divided into several clusters. As a result, the cluster's member nodes can avoid the energy consumed by forwarding [12]. An assistant node was introduced to separate data forwarding from transmission, addressing the premature energy depletion of certain nodes caused by repeated data forwarding. These approaches, however, have several shortcomings. First, they rely on exact, energy-intensive mathematical models to create and are highly difficult to plan. Second, those approaches are not flexible enough to adapt to varied network topologies and scales, which renders WSNs unfeasible because of congestion. As a result, new approaches must solve these issues [13].

Recent advancements in machine learning (ML) techniques have addressed the limitations of traditional routing protocols in WSNs. ML offers a flexible and adaptive approach capable of handling the complex data and computational demands associated with designing effective routing algorithms for WSNs [14].

The inadequacies of traditional clustering and routing methods for WSNs that lead to the necessity of integrating machine learning algorithms with clustering and routing include:

1) Limited Scalability and Flexibility: Simple clustering and routing methods may not handle large-scale WSNs, as they may struggle to manage the complexity and resource requirements of the network. Ineffective routing and decreased network performance can result from basic clustering and routing techniques that are not sensitive to changes in the network, such as node failures, mobility, or environmental factors [15].
2) Limited Data Aggregation: Ineffective use of data aggregation techniques by simple clustering and routing algorithms may cause needless data transfer and higher energy costs.
3) Energy Inefficiency: These techniques might not properly optimize energy consumption since they might not consider network load, node proximity, or available energy when choosing routing paths [16].

To enhance the self-reliance of WSNs, machine learning algorithms are being explored for tasks such as automatic clustering, cluster head selection, routing path discovery, and data aggregation [17]. These algorithms hold the potential to significantly improve network lifetime by reducing power consumption and optimizing routing path selection. This research focuses on two key challenges: cluster formation and routing path selection. The effectiveness of the proposed approach is evaluated based on three key performance indicators: improved energy efficiency, optimized routing paths, and extended network lifetime. We use K-means for cluster formation and a Q-learning algorithm for routing path selection. In multi-hop topology, there can be multiple paths for data routing. If one path fails, the self-organization and dynamic topology quality make it possible to select a different route to make the network operational.

A. RESEARCH CONTRIBUTION AND IMPACT ON WSN OPTIMIZATION

Our study is the first to explore the combined use of Q-Learning and K-Means clustering to enhance both clustering and routing mechanisms in WSNs. This innovative approach addresses several persistent challenges in WSNs, such as limited scalability, flexibility, inefficient data aggregation, and high energy consumption. By integrating these two Machine Learning techniques, we have developed a comprehensive solution that optimizes clustering and routing, a departure from traditional methods that typically handle these tasks separately or rely on a single algorithm for both.

Our research demonstrates the practical advantages of this combined approach and sets a new standard for achieving greater energy efficiency and network durability in WSNs. Through extensive comparisons with established protocols such as LEACH and its multi-hop variants DMHT LEACH and EDMHT LEACH, our study validates the effectiveness of this integrated method across various metrics, including the number of alive nodes, average energy consumption, remaining energy levels, and packet reception after 600 rounds in networks comprising 400 nodes [18], [19], [20]. The results of our research have been compared with the well-established LEACH algorithm and its two multi-hop variants, DMHT LEACH and EDMHT LEACH, to validate the effectiveness of our approach. The comparison was conducted across multiple critical performance metrics, including the number of alive nodes, average energy consumption, remaining energy levels, and the number of packets received after 600 rounds in a network of 400 nodes.

The research is conducted to meet the following main objectives:

OBJ 1: The principal objective is to optimize cluster formation and routing path selection in WSNs using machine learning techniques, particularly Q-learning and K-means, to increase energy efficiency and data transmission efficacy.

OBJ 2: To validate the proposed EEMLCR method's efficacy in enhancing clustering and routing in WSNs, a comparison with other algorithms, such as LEACH, DMHT LEACH, and EDMHT LEACH, is conducted.

OBJ 3: The main goal is to enhance energy-efficient WSNs by incorporating machine learning algorithms into the routing and clustering procedures to achieve a more efficient and sustainable network operation.

II. LITERATURE REVIEW

Many researchers have presented the routing and clustering techniques in WSNs [21]. Below are some of the studies, including the techniques employed and their significant features:

The routing techniques in WSNs are classified as flat, hierarchical, and location-based [22]. These protocols are QoS-based, query-based, non-coherence, coherence-based, negotiation-based, and Multipath-based [23]. One of the main design goals of WSNs is to communicate data in an energy-efficient manner. Some routing challenges in WSNs are node deployment, efficient energy consumption without losing accuracy, data reporting, node/link heterogeneity, fault tolerance, scalability, network dynamics, transmission media, connectivity, coverage, data aggregation, and quality of service. Flat routing is a category of routing protocol in which every node plays the same role (sensing task). Hierarchical routing can also be called cluster-based routing, which works energy-efficiently. In this type of routing, high-energy nodes become cluster heads and perform processing and data transmission tasks. In location-based routing, sensor nodes are addressed by location via GPS, and energy efficiency is ensured by applying a sleep-wake cycle. Further routing protocols are divided into categories based on protocol operation, i.e., Multipath (uses alternative paths), query-based (receiver queries for data), negotiation-based (use messages to remove redundant data), QoS-based (aims to balance data quality and energy consumption), non-coherence (node's process raw data locally) & coherence-based routing (data is forwarded to aggregators after time stamping and duplicate suppression).

The challenges and opportunities in WSNs using machine learning techniques are discussed in [24]. This work studies efficient, robust communication protocols to deal with energy consumption challenges and enable network operations with prolonged network lifetime. The machine-learning algorithm is based on various techniques, such as supervised, unsupervised, reinforcement, and evolutionary computing algorithms. Unlike supervised learning, unsupervised learning processes unlabeled datasets without guidance to solve more complex problems. Unsupervised learning solves problems like data aggregation, connectivity, clustering, and routing. The most popular and simple unsupervised learning algorithm is k means, which divides data points into k clusters by creating a centroid for each cluster. The reinforcement-learning algorithms learn from the environment with no training data set. In reinforcement learning, agents select actions based on the potential rewards they expect to receive. Actions associated with higher rewards are more likely to be chosen by the agent. Q-learning is an example of reinforcement learning. In WSNs, energy is a constraint that makes it difficult to send all the sensed data to the base station in an energy-efficient manner. As machine learning techniques can process massive data quickly, use energy efficiently, and provide higher accuracy, they can be employed to overcome wireless sensor network problems stated above.

The study conducted by [25] provides a comprehensive examination of various Energy Efficient Routing Protocols (E-ER-Ps) designed specifically for Underwater WSNs (UWSNs). A comparative analysis between contemporary machine learning (ML) methodologies and traditional protocols, given that ML-based strategies have demonstrated considerable promise in addressing the complex challenges that UWSNs encounter. This manuscript aspires to review distinct E-ER-Ps from multiple perspectives pertinent to UWSNs critically. The study proposed an innovative taxonomy for their systematic classification to facilitate a deeper understanding of the architecture and applications of these protocols. While ML-based protocols are assessed regarding their adaptability, predictive capabilities, and overall advancements in efficiency, conventional protocols are scrutinized based on their routing strategies and enhancements in energy efficiency. An exhaustive comparative analysis elucidates various protocols' merits, demerits, and potential applications. Furthermore, a critical examination of the role of ML, incorporating intelligent and adaptive routing strategies, is presented, underscoring the technology's capacity to fundamentally transform the management of UWSNs. To devise and implement E-ER-Ps for UWSNs, the study concluded by emphasizing the extant challenges, including the requisite for real-time adaptability, resilience to environmental variations, and compatibility with pre-existing network infrastructures. The advancement of ML-based methodologies, hybrid strategies that amalgamate traditional and ML-based approaches, and the design of protocols capable of dynamically adapting to the evolving conditions of underwater ecosystems are underscored as future research priorities.

The research [26] introduces an energy-efficient clustering methodology that employs gray prediction techniques. The primary objective is to enhance energy efficiency within wireless sensor networks (WSNs). A dual-end data prediction mechanism is implemented to mitigate data redundancy. An energy-distance factor is utilized to identify optimal cluster heads, thereby facilitating effective load balancing. Comprehensive simulations indicate that the Energy Clustering with Gray Prediction (ECGP) surpasses the performance of existing clustering algorithms. ECGP considerably extends the network's operational lifespan while decreasing energy consumption. An energy-efficient clustering methodology utilizing gray prediction is hereby proposed. The theoretical groundwork of the Dual-end Data Prediction Mechanism (DDPM) is elucidated. Specifics regarding the Energy Distance Factor (EDF) and the Dynamic Time Cluster Consensus (DTCC) are thoroughly examined. The Least Squares Method is employed to estimate parameters. The First Order Accumulating Generation Operator is applied to the processing of data. The predetermined error has implications on both prediction accuracy and energy consumption. Outliers captured by sensor nodes are considered to be of minimal significance. Energy consumption escalates as the level of prediction accuracy increases.

An energy-efficient hierarchical cluster-based routing survey was conducted by [27]. This study is focused on energy-efficient hierarchical cluster-based routing. The main objective of WSNs is to produce information from raw local data obtained by sensor nodes. The limited power of WSNs mandates the design of an energy-efficient communication protocol. Hierarchical clustering is a distributed randomized clustering algorithm that maximizes the network's lifetime.

The conducted research [28] introduces a routing protocol ''Data Synchronized-Machine Learning (DS-ML)'' designed for Internet of Things (IoT) enabled WSNs. DS-ML facilitates the optimization of power consumption within wireless IoT devices. It demonstrates an ability to adapt the variations and disturbances within the network. The approach integrates methodologies derived from Reinforcement Learning and Meta-Learning paradigms. DS-ML enhances energy efficiency while concurrently minimizing data transmission latency. The framework bolsters adaptability across a spectrum of network conditions. The Software-Defined Networking (SDN) architecture endorses the incorporation of optimization strategies. The manuscript systematically tackles the challenges inherent in IoT communication services. It emphasizes the importance of energy efficiency within wireless sensor networks by utilizes a Reinforcement Learning-Driven (RL) algorithm. It incorporates Meta-Learning to augment adaptability. The DS-ML framework enhances routing efficiencies within Software-Defined Networking (SDN) environments. It employs trial-and-error methodologies to ascertain optimal strategies. The algorithm accumulates data regarding network topology and performance metrics. It delineates state space, action space, and reward functions.

The research conducted by [29] covers routing techniques, cluster head selection, and energy consumption. The goal of the study is to extend the network lifetime and residual energy of nodes. Both an election process and a clustering mechanism are suggested. For efficiency, the protocol reduces the distance to the base station. To lessen energy transfer, it uses a data fusion process. The study emphasizes developments in sensor network applications and technologies by utilizing the I-LEACH technique for routing and data clustering. The protocol seeks to increase the network lifetime and residual energy of nodes. It transfers data to the Base Station via a multi-hop technique. The research utilized the Cluster Head election procedure to cluster the data. Energy models are used to prevent unnecessary energy transfer, but node energy consumption remains a significant issue.

Khan and Samad studied the application of machine-learning algorithms in WSNs [30]. The study investigates the design issues in WSNs and suggests that machine-learning strategies can fix many issues. Rezaei et al. performed a survey and research analysis on WSNs [31]. The authors reviewed the clustering techniques of WSNs and extracted 215 of the most important techniques. These techniques were further reviewed and categorized based on the clustering objectives. This work mentions various clustering objectives, such as energy consumption, fault tolerance, load balancing, QoS, scalability, and solutions like CH rotation, balanced clusters, replastering, and reduced overhead.

Table 2 shows a comparison of the literature related to this work. The classification of our literature review is based on unsupervised learning, reinforcement learning, and hybrid techniques. Most of the literature review focuses on energy and routing efficiency.

Under the Unsupervised Learning category, five techniques were reviewed: OPSKC, IS-K means, KEAC, MRRCE, and Energy Efficient Routing using Mobile Sink. OPSKC uses a fixed packet size to increase network lifetime and aims for energy efficiency. IS-K utilizes multiple cluster heads, data sending schedules, and next CH activation based on energy threshold to improve energy efficiency. KEAC aims to achieve uniform/balanced CHs and clusters and network load balancing. Energy Efficient Routing with Mobile Sink divides the network into sectors, uses a mobile sink, and selects CHs based on energy and distance to improve energy efficiency.

The reinforcement Learning category includes five reviewed techniques: Q learning, QDAEER, Q learning-based routing in WSN, QDAR, and geo-casting based on Q learning. Q learning-based techniques use reward-based routing, switch between explore and exploit, and implement a greedy strategy to improve energy and routing efficiency. QDAEER uses defined sensing schedules to achieve energy and routing efficiency. Q learning-based routing in WSN routes data is used only when required to improve energy and routing efficiency. QDAR uses a sleep-wake cycle to improve energy efficiency. Geo-casting based on Q-learning uses relay nodes and the Fermat point to get relay nodes to improve energy and routing efficiency.

The Hybrid Technique category includes four reviewed techniques: K means++-based clustering and routing, enhanced K means with Disjkstra, Q probabilistic routing, DMHT LEACH, and EDMHT LEACH. K means++-based clustering and routing uses the shortest path for intercluster communication and CH rotation based on energy levels to achieve energy efficiency. Enhanced K-means with Disjkstra removes data redundancy. Q probabilistic routing uses Q-learning to calculate routing costs based on previous traffic patterns to achieve energy and routing efficiency. DMHT LEACH and EDMHT LEACH both use LEACH for clustering, and intermediate nodes facilitate data transmission, while DMHT LEACH uses multi-hop routing and changes routes based on CH energy and distance. EDMHT LEACH improves cluster formation by checking node energy before selecting.

III. METHODOLOGY

Apart from hardware (production of low-cost and tiny sensor nodes) [32], achieving good coverage while maintaining or preserving energy [33], clustering and routing path selection

TABLE 2. Comparison of techniques implemented by using unsupervised, supervised and reinforcement learning.

Classification	Reviewed Techniques	Characteristics	Goal
Unsupervised Learning	OPSKC	Fixed packet size to increase network lifetime.	Energy Efficiency
	IS-K means	Multiple cluster heads. Data sending schedules. Next CH activates when energy falls below threshold.	Energy Efficiency
	KEAC	Uniform/balanced CHs and clusters. Network load balancing.	Energy Efficiency
	Energy Efficient Routing with Mobile Sink	Network divided in sectors. Mobile sink. CH selection based on energy and distance.	Energy Efficiency
	MRRCE	CH selection improved using Steiner points.	Energy
Reinforcement Learning	Q learning	Reward-based routing. Lower transfer delay = Higher reward. Switch between explore and exploit. Uses greedy strategy.	Energy + Routing Efficiency
	QDAEER	Q learning-based routing. Defined sensing schedules.	Energy + Routing Efficiency
	Q-learning Based Routing in WSN	Uses Q learning-based routing. Data is routed only when required.	Energy + Routing Efficiency
	QDAR	Q-learning used to route data. Sleep-wake cycle implemented	Energy + Routing Efficiency
	Geocaching based on Q learning	Reward-based routing, relay nodes. Fermat point used to get relay nodes. Geocaching technique to transmit query packets.	Energy + Routing Efficiency
Hybrid Techniques	K means++ based clustering and routing	Address uneven clustering. Shortest path for intercluster comm. CH rotation based on energy levels.	Energy Efficiency
	Enhanced K means with Disjkstra	Enhanced k means with Dijkstra. Removes data redundancy.	Removes Data Redundancy
	Q Probabilistic Routing	Q-learning is used. Routing cost calculated by previous traffic patterns.	Energy + Routing Efficiency
	DMHT LEACH	Multi hop routing with LEACH for clustering. Changes routes based on CH energy and distance. TDMA schedules for transmission.	Energy + Routing Efficiency
	EDMHT LEACH	Improved cluster formation. Node energy checked before selecting as CH. Intermediate nodes facilitate data transmission.	Energy + Routing Efficiency

are known challenges in WSNs. Two machine learning algorithms, K means (clustering) and Q-learning (routing path selection), are implemented to address these challenges. The results are validated based on improved energy efficiency, routing path selection, and network lifetime. The proposed method is compared with the LEACH protocol and two dynamic multi-hop variants of LEACH (DMHT and EDMHT technique). The proposed and existing methods are implemented in MATLAB for simulation and result validations.

A. PROPOSED METHOD FOR CLUSTERING

K-means clustering offers several advantages, including its widespread popularity, straightforward implementation on large datasets, integration with other methods to extend the lifetime of wireless networks, and guarantee that it assigns all data points to a cluster [34]. However, a key limitation of K-means is its sensitivity to the initial cluster centroids, which can be mitigated through manual selection. Q-learning, on the other hand, is a model-free reinforcement learning technique [35]. Unlike model-based approaches, Q-learning does not require a predefined model of the environment, leading to potentially lower energy consumption. It operates in an off-policy manner, prioritizing actions with the highest estimated future reward (Q-value).

In contrast, SARSA, another model-free reinforcement learning algorithm, updates the Q-value based on the next state's Q-value and the chosen action, leading to a generally slower learning process. We used K-means and Q-learning algorithms because of their computational efficiency and ability to handle large datasets, which is crucial for the scalable WSNs of the K-means algorithm. Q-learning was chosen for its adaptability to dynamic environments and ability to optimize long-term rewards, addressing the energy constraints in WSNs.

1) K-MEANS ALGORITHM

The K-means algorithm is a popular unsupervised machine-learning technique [36]. It randomly selects initial centroids (cluster centers) from the input data. Each iteration (round) of the algorithm refines the clusters based on the centroid selection. A key challenge of k-means is that random selection of centroids can lead to unbalanced or even empty clusters. K-means takes data points and a desired number of clusters as input and outputs a set of clusters. This research proposes a method for selecting and rotating cluster heads in WSNs. This method considers the residual energy of sensor nodes and their minimum distance to the base station. The algorithm steps are given in Figure 2 below. Figure 3 provides the flow chart for the process of formation of clusters in the K-means clustering algorithm. Figure 4 presents the pseudocode for forming clusters and selecting cluster heads using the K-means method.

Here is a refined description of the algorithm given, as shown in Figure 2:

1. Initialization: Initially, K cluster centers are chosen. This can be done by randomly selecting K data points from the dataset or utilizing domain-specific knowledge or heuristics to determine the initial centers.

2. Assigning Data Points to Clusters: For each data point, the Euclidean distance is calculated between the point and each cluster center. Subsequently, the data point is assigned to the cluster with the closest center.

3. Recalculating Cluster Centers: After assigning all data points to clusters, the cluster centers are recalculated by computing the mean of all data points assigned to each cluster. This new mean position serves as the updated centroid for the cluster.

4. Repeating Steps 2 and 3 Until Convergence: Steps 2 and 3 are iteratively repeated until convergence is achieved. Convergence occurs when the cluster centers no longer undergo significant changes or when a predetermined number of iterations is reached. During each iteration, data points are reassigned to their nearest cluster center, and centroids are recalculated accordingly.

5. Outputting the Final Clusters: Upon convergence, the final clusters are output. Each cluster comprises the data points closest to its centroid, thereby forming well-defined clusters representative of the input data distribution.

The K-means algorithm converged when the change in cluster centroids between consecutive iterations was less than a predefined threshold of 0.001 or after a maximum of 100 iterations. This ensured both accuracy and computational efficiency. While K-means is sensitive to initial centroid placement, we mitigate this through multiple initializations. DBSCAN was also explored, which showed promise in irregular network topologies but had higher computational costs.

Input:
- Data: Set of data points
- k: Number of clusters
- MaxIterations: Maximum number of iterations

Initialization:
1. Randomly select k data points from the dataset as initial cluster centroids.

Main Loop:
2. Repeat until convergence or maximum iterations reached:
a. Assign each data point to the nearest cluster centroid:
For each data point d in Data:
Calculate the Euclidean distance between d and each centroid.
Assign d to the cluster with the nearest centroid.

b. Update cluster centroids:
For each cluster:
Calculate the mean of all data points assigned to the cluster.
Update the cluster centroid to be the mean.

c. Check for convergence:
If the cluster centroids do not change significantly or maximum iterations reached, exit the loop.

Output:
3. Return the final cluster assignments and centroids.

FIGURE 2. The key step in k-means algorithm.

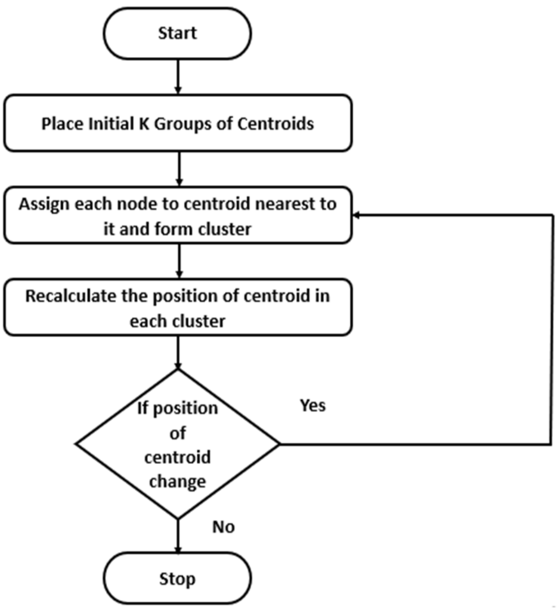

FIGURE 3. Flow chart of k-means algorithm.

2) Q-LEARNING ALGORITHM

The Q-learning technique uses a reward-based system to identify the optimal action based on sensor data. It prioritizes actions with the highest Q-value, employing a greedy

```
Input:
- A set of sensor nodes S = {S1, S2, S3, ..., Sn}
- Number of clusters k (k > 1)

Output:
- A set of k clusters generated

Function EuclideanDistance (node1, node2):
   Calculate the Euclidean distance between node1 and node2
   Return the distance

Function FindClosestCentroid (node, centroids):
   min_distance = Infinity
   closest_centroid = None
   For each centroid in centroids:
      distance = EuclideanDistance (node, centroid)
      If distance < min_distance:
         min_distance = distance
         closest_centroid = centroid
   Return closest_centroid

Function RecalculateCentroids(clusters):
   For each cluster in clusters:
      If cluster is not empty:
         Calculate the mean of all data points (nodes) in the cluster
         Update the centroid position to the mean

Function InitializeCentroids (k, S):
   Randomly select k nodes from S as initial centroids
   Return the initial centroids

Function ClusterHeadSelectionAndRotation (S, k):
   centroids = InitializeCentroids (k, S)
   Repeat:
      clusters = {} # Dictionary to store clusters
      For each node in S:
         closest_centroid = FindClosestCentroid (node, centroids)
         Add node to the cluster of closest_centroid
      old_centroids = centroids
      RecalculateCentroids(clusters)
      If centroids have not changed significantly:
         Break the loop
   Until convergence or predefined number of iterations
   Return clusters

clusters = ClusterHeadSelectionAndRotation (S, k)
Output clusters
```

FIGURE 4. **Pseudocode for clustering k-means [37].**

approach. However, this strategy can lead to convergence issues. The Q-routing protocol, designed to estimate end-to-end delay, selects the sensor node (SN) with the lowest Q-value for packet forwarding. While this method aims to minimize delay, it requires multiple attempts to converge and may suffer from outdated estimations due to the dynamic nature of WSNs. Figure 5 shows the routing in a multi-hop WSN using a Q-learning-based routing protocol.

The reward function was designed to balance energy efficiency and path optimality. It was defined as:

$$R = \alpha(1/E) + \beta(1/H) \tag{1}$$

where E is the energy consumed along the path, H is the hop count, and α and β are weighting factors set to 0.6 and 0.4 respectively, prioritizing energy efficiency slightly over path length.

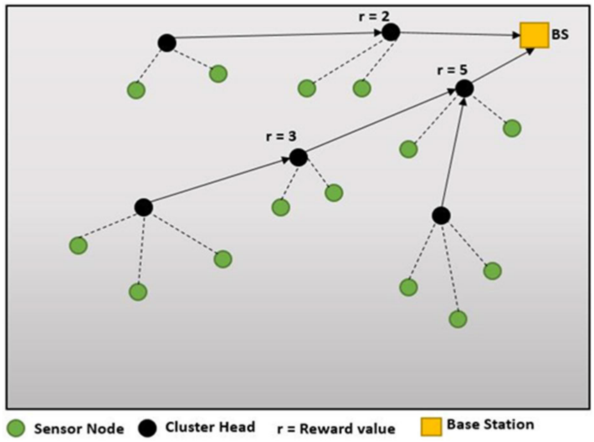

FIGURE 5. **Q learning based routing techniques.**

```
Initialize Q (s, a) arbitrarily
Repeat (for each episode)
      Initialize s
      Repeat (for each step of episode):
            Choose a from s using policy derived from Q
      Take action a, observe r, s'
      Q (s, a) = Q (s, a) + alpha * (r + gamma * max (Q (s', a'))
– Q (s, a))
      Where s is the current state, a is the chosen action, s' is the
      new state, r is the reward, alpha is the learning rate, and
      gamma is the discount factor
Update the state
```

FIGURE 6. **Pseudocode for Q learning.**

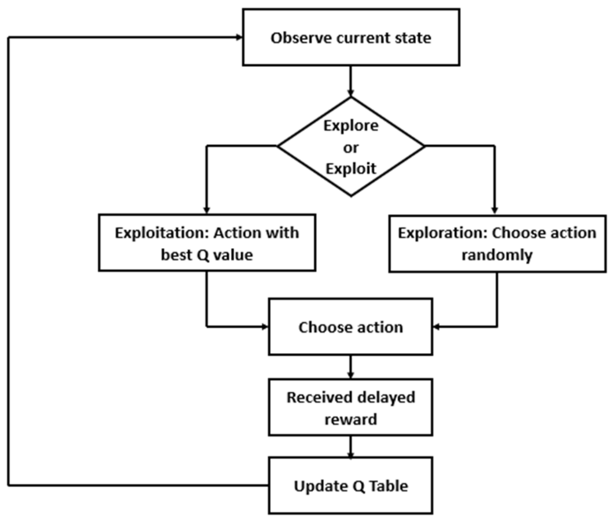

FIGURE 7. **Flow chart of Q learning.**

Step 1. Define the State Space: The state space represents all possible states the agent can encounter, influenced by the problem and environment.

Step 2. Define the Action Space: The action space encompasses all feasible actions the agent can undertake in each state, determined by the problem and environment.

Step 1: Define the state space
Step 2: Define the action space
Step 3: Define the reward function
Step 4: Initialize the Q-table with zeros
Step 5: Repeat for each episode:
a. Choose an initial state
b. Repeat until the episode terminates:
i. Choose an action using an exploration/exploitation strategy (e.g., epsilon-greedy)
ii. Execute the chosen action
iii. Observe the new state and the reward
iv. Update the Q-value for the (state, action) pair using the Q-learning equation:
Q (s, a) = Q (s, a) + alpha * (r + gamma * max (Q (s', a')) – Q (s, a))
Where s is the current state, a is the chosen action, s' is the new state, r is the reward, alpha is the learning rate, and gamma is the discount factor
c. Update the state
Step 6: Use the learned Q-table to make decisions in real-time

FIGURE 8. Pseudocode for routing path selection (Q-learning [37]).

Step 3. Define the Reward Function: The reward function computes a scalar reward signal based on the current state and action, guiding the agent to maximize cumulative reward over time. In this context, reward reflects transfer delay estimates, with lower delays yielding higher rewards.

1. **Initialize the Q-table:** The Q-table maps each (state, action) pair to a Q-value, indicating the expected cumulative reward if the agent chooses that action in that state and acts optimally thereafter. Initially, the Q-table is initialized with zeros.
2. **Repeat for each episode:**

a. Choose an initial state: The agent begins in an initial state.

b. Repeat until the episode terminates:

i. Choose an Action: Employing an exploration/ exploitation strategy, the agent selects an action in the current state, balancing trying new actions and exploiting known optimal ones.

ii. Execute the Action: Employing an exploration/exploitation strategy, the agent selects an action in the current state, balancing between trying new actions and exploiting known optimal actions.

iii. Observe New State and Reward: The agent observes the new state and the reward from executing the chosen action.

iv. Update the Q-Value: Using the Q-learning equation, the agent updates the Q-value for the current (state, action) pair, considering the current and expected future rewards.

c. Update State: The agent transitions to the new state observed in the previous step.

Step 4. Utilize the Learned Q-Table: Once the agent completes multiple episodes and updates its Q-table, it can make real-time decisions. By referencing the Q-values for the current state, the agent selects the action associated with the highest Q-value.

1. Initialization:
- K = number of clusters
- X = data points representing sensor nodes in WSN
- S = state space representing possible paths
- A = action space representing possible next hops
- R = reward function (energy efficiency, latency, reliability)
- alpha = learning rate for Q-learning
- gamma = discount factor for Q-learning
- epsilon = exploration rate for Q-learning
- max_iterations = maximum number of iterations for K-means clustering
- max_episodes = maximum number of episodes for Q-learning

2. K-means Clustering:
- Initialize cluster centers randomly
- Repeat for max_iterations:
  - Assign data points to the nearest cluster center
  - Update cluster centers to the mean of assigned points
  - Select cluster heads based on remaining energy level and distance to the base station

3. Route Selection:
- For each node in the path:
  - Check if the node is a Cluster Head (CH)
  - If CH:
    - Determine a valid path to the next node
  - If not CH:
    - Assign an invalid path

4. Q-learning:
- Initialize Q-values arbitrarily
- Repeat for max_episodes:
  - Select an initial state (source node)
  - While the current state is not the destination node:
    - With probability epsilon, select a random action (next hop). Otherwise, select the action with the highest Q-value for the current state.
    - Take the selected action, observe the reward based on energy efficiency, latency, and reliability.
    - Update the Q-value using the Q-learning equation:
      Q(s, a) = Q(s, a) + alpha * (r + gamma * max(Q(s', a')) – Q(s, a))
    - Set the current state to the next state (next hop)

5. Cluster Head Rotation:
- Rotate cluster heads based on remaining energy level and distance to the base station

FIGURE 9. Pseudocode for the proposed technique of clustering and routing path selection (EEMLCR).

Figure 8 provides the pseudocode for implementing the Q-learning method to select optimal routes in a multi-hop communication-based WSN. Figure 9 provides the pseudocode for our proposed method of K-means-based clustering and Q-learning-based routing in WSN.

3) PROPOSED EEMLCR

In this proposed technique, two machine learning-based algorithms are combined to address two main problems of WSNs: clustering and routing. The pseudocode for the proposed EEMLCR is given in Figure 9, and the algorithm breakdown of EENLCR is given in Figure 10.

4) ASSUMPTIONS

The following assumptions are made regarding the network model for the proposed method.

1) All nodes are static.

1. Initialization: Deploy sensor nodes and set initial energy levels
2. Clustering Phase:
a. Apply K-means clustering to group nodes based on spatial proximity
b. Select cluster heads based on residual energy and centrality within clusters
3. Routing Phase:
a. Initialize Q-values for all possible state-action pairs
b. For each data transmission:
i. Select next-hop node using ε-greedy policy
ii. Transmit data and observe energy consumption
iii. Update Q-values using the Q-learning update rule
4. Maintenance:
a. Periodically re-cluster nodes to balance energy consumption
b. Update cluster head selection based on current energy levels
5. Repeat steps 3-4 until network lifetime ends

FIGURE 10. Algorithm breakdown of EEMLCR.

2) Sensors are homogenous with the same initial energy.
3) Only one static base station is far from the sensing field.
4) The base station knows the geographical location of sensor nodes.
5) All CHs send aggregated data to the base station.
6) The security of the network is addressed.

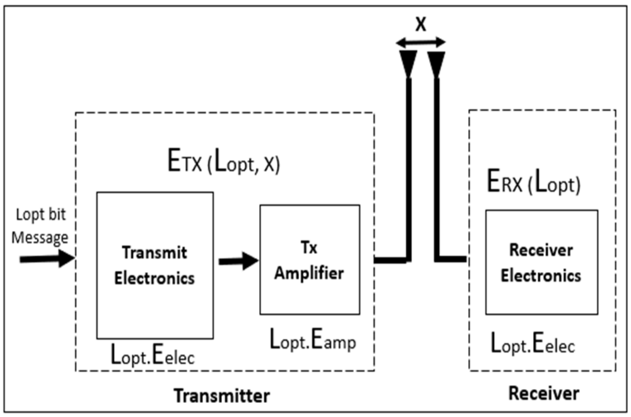

FIGURE 11. Wireless sensor network energy model.

5) ENERGY MODEL

Figure 11 shows energy dissipation model as mentioned in [38].

In this study, we employ the First-Order Radio Energy Model, which is widely used in WSNs to simulate energy dissipation in data transmission and reception. The model accounts for transmission, reception, and amplification energy consumption based on distance and data size.

Assumptions of the Energy Model: The assumptions of the energy model used in this study are as follows:

- Each sensor node starts with an initial energy level of E_initial.
- Nodes consume energy for data packet transmission (ETX) and reception (ERX).
- The communication model assumes a free-space (d^2) or multi-path fading (d4) channel, depending on the distance d between the sender and receiver.
- Data aggregation and processing energy are negligible in comparison to transmission energy.

Energy Consumption Formulations: The energy consumption in WSNs is modeled using the following equations:

1) **Transmission Energy Consumption:** The energy required to transmit L-bit data over a distance d is:

$$E_{TX}(d) = \begin{cases} L \times E_{elec} + L \times \varepsilon_{fs} \times d^2,\ d < d_0\ (free-spacemodel) \\ L \times E_{elec} + L \times \varepsilon_{mp} \times d^4,\ d \geq d_0\ (multi-pathmodel) \end{cases} \tag{2}$$

where:

- E_{elec} = 50 nJ/bit (Energy for running transmitter/receiver electronics)
- ε_{fs} = 10 pJ/bit/m^2 (Energy for free-space model)
- ε_{mp} = 0.0013 pJ/bit/m^4 (Energy for multi-path model
- d_0 = Threshold distance calculated as:

$$d_0 = \sqrt{\frac{\varepsilon_{fs}}{\varepsilon_{mp}}} \tag{3}$$

2) **Reception Energy Consumption:** The energy consumed to receive an L-bit packet is:

$$E_{RX} = L \times E_{elec} \tag{4}$$

3) **Cluster Head Energy Dissipation:** The CHs perform additional tasks such as data aggregation and forwarding. The energy required for data aggregation is:

$$E_{DA} = L \times E_{agg} \tag{5}$$

where E_{agg} = 5 nJ/bit (Energy for data aggregation)
The total energy consumed by a CH is:

$$E_{CH} = E_{TX} + E_{DA} + \sum E_{RX} \tag{6}$$

Energy Consumption (Transmission): The equation given as (7) is used to calculate the energy consumption at transmission end.

$$\mathrm{E_{TX}(L_{opt}, x) = L_{opt} * E_{elec} + L_{opt}} * \varepsilon_{\mathrm{amp}} \tag{7}$$

where:

- E_{TX}: Expected Transmission Count
- E_{RX}: Expected Receiving Count
- L_{opt}: Optimal path length between the sender and receiver nodes
- x: The receiving node
- E_{elec}: Energy required to run the transmitter or receiver electronics circuitry
- ε_{amp}: Energy required to transmit a single bit of data over a unit distance

Equation 7 shows energy consumption on the transmitter end. The formula tells us that the expected transmission count (E_{TX}) is the product of the optimal path length (L_{opt}) and the sum of the energy required to transmit data over that path. The energy cost of transmitting data consists of two parts: the energy required to operate and the electronic circuitry (E_{elec}), and the energy required to transmit data over the wireless medium (ε_{amp}), whereas x is the receiving node. In general, Equation 1 is used to estimate the energy cost of transmitting data over a wireless sensor network, which is important in designing efficient and effective routing protocols for the network. Minimizing the ETX can reduce the network's energy consumption and increase its lifetime.

Energy Consumption (Receiving): The equation given as (8) is used to calculate the energy consumption at receiving end.

$$E_{RX}(L_{opt},) = L_{opt} * E_{elec} \tag{8}$$

Equation (8) shows that the expected reception count (E_{RX}) is the product of the optimal path length (L_{opt}) and the energy required to operate the receiver electronics circuitry (E_{elec}).

This means that the energy cost of receiving data is directly proportional to the distance between the sender and receiver nodes. In general, equation 2 is used to estimate the energy cost of receiving data over a wireless sensor network, which is also an important factor in designing efficient and effective routing protocols for the network. Minimizing the ERX can reduce the network's energy consumption and increase its lifetime.

6) INTEGRATION OF K-MEANS AND Q-LEARNING

The following steps outline how K-means and Q-learning are integrated:

Step 1: Initialization of the WSN Network Model

- A static WSN topology is defined, with N sensor nodes deployed in a 400 × 400 m^2 sensing region.
- Each node is initialized with 10 Joules of initial energy.
- The Base Station (BS) is positioned outside the sensing area at coordinates (410, 410).
- Each node starts with an equal probability of being selected as a Cluster Head (CH).

Step 2: Cluster Formation Using K-Means Algorithm

The clustering process is carried out using the K-means algorithm through the following steps:

1) **Random selection of initial cluster centroids:** The algorithm randomly selects K initial centroids from the sensor node locations.
2) **Assignment of sensor nodes to the nearest cluster centroid:** The Euclidean distance between each sensor node and each centroid (d_(i,j)) is calculated as:

$$d\_(i,j) = \sqrt{}((x\_i{-}x\_j) \wedge 2 - (y\_i - [xy]\_j) \wedge 2) \tag{9}$$

3) **Updating cluster centroids:** The centroid positions are recalculated based on the mean of all nodes assigned to the cluster.
4) **Cluster Head Selection:** The node with the highest residual energy in each cluster is selected as CH.
5) **Cluster Head Rotation:** CHs are rotated periodically to ensure balanced energy consumption.

Step 3: Routing Path Selection Using Q-Learning: After clusters are formed, Q-learning is applied to select optimal routing paths:

Defining Q-learning components:

- **State (S):** Each sensor node represents a state.
- **Action (A):** The set of neighboring nodes to which data can be transmitted.
- **Reward Function (R):** reward function is defined as:

$$R = \alpha(1/E) + \beta(1/H) \tag{10}$$

- where E is the remaining energy of the node, H is the hop count, and α and β are weighting factors (set to 0.6 and 0.4).
- **Initializing Q-Table:** The Q-table is initialized to store Q-values for each (state, action) pair.
- **Choosing the Best Next-Hop Node:** Using an ε-greedy policy, a sensor node either:
- **Exploits** (chooses the path with the highest Q-value), or
- **Explores** (randomly selects a new path with probability ε).
- **Updating Q-Values:** After selecting a next-hop node and receiving feedback, the Q-value is updated using the following equation:

$$Q(s,a) = Q(s,a) + \alpha\left[R(s,a) + \gamma \max_{\acute{a}} Q(\acute{s},\acute{a}) - Q(s,a)\right] \tag{11}$$

where:
γ is the discount factor (set to 0.9 to prioritize long-term rewards) and $Q(\acute{s},\acute{a})$ is the maximum Q-value of the next state.

- **Packet Forwarding:** The node with the highest Q-value in its routing table is selected as the next-hop node.
- **Termination Condition:** The process repeats until the data reaches the Base Station (BS).

7) JUSTIFICATION OF USING K-MEANS AND Q-LEARNING CLUSTERING ALGORITHM

Clustering is a critical step in WSNs to group sensor nodes, reduce energy consumption, and improve network lifetime. Various clustering algorithms exist, such as LEACH, Fuzzy C-Means (FCM), and Particle Swarm Optimization (PSO). However, we selected K-means clustering due to the following advantages [39], [40], [41]:

1) **Computational Efficiency:** K-means has a computational complexity of O(NK), making it faster and scalable for large WSNs compared to FCM ($O(N^2K)$) or PSO (which involves additional fitness function evaluations). In contrast, LEACH selects CHs randomly, leading to suboptimal clustering.

2) **Energy Balancing:** K-means ensures uniform cluster distribution, reducing the overload on any single node. LEACH often results in uneven cluster sizes, leading to premature energy depletion in some areas.
3) **Reduced Control Overhead:** Unlike PSO or Genetic Algorithms, which require iterative optimization steps, K-means has a lower control message overhead, making it more suitable for real-time WSN applications.

The justifications for using the Q-learning are as follows [42], [43], [44], [45], [46]:

1) **Adaptive Path Selection:** Unlike fixed shortest-path routing, Q-learning dynamically learns and adapts to changes in energy levels and network topology. AODV and DSR do not consider residual energy, leading to rapid node failures.
2) **Energy-Aware Decision Making:** Q-learning assigns Q-values to different routes based on energy availability and hop count, ensuring that high-energy paths are preferred. Traditional routing protocols do not optimize for energy consumption.
3) **Scalability and Robustness:** Q-learning performs better in dynamic WSNs, as it does not rely on static tables like Dijkstra's algorithm. It can self-adjust based on network congestion and node failures.

Table 3 gives a summary of justifications given above.

TABLE 3. Summary of algorithms justifications.

Algorithm	Alternative Methods	Limitations of Alternatives	Justification of Our Choice
K-Means (Clustering)	LEACH, Fuzzy C-Means, PSO	LEACH: Random CH selection. FCM: High computational cost. PSO: High control overhead.	K-means ensures uniform cluster distribution, is computationally efficient, and balances energy consumption.
Q-Learning (Routing)	AODV, DSR, Dijkstra	AODV & DSR: Not energy-aware. Dijkstra: Static routing.	Q-learning dynamically adjusts routes based on energy levels and minimizes energy consumption.

IV. RESULTS

This section analyzes and presents the results of the proposed method EEMLCR and their comparison with LEACH and its multi-hop variants DMHT and EDMHT LEACH. According to the results, EEMLCR performs better or at the same level as the other techniques. The EEMLCR shows better scalability potentials, providing better average energy consumption than LEACH, DMHT LEACH, and EDMHT LEACH. Concerning the number of alive nodes and throughout, EEMLCR performs better or the same as DMHT and EDMHT LEACH.

A. EXPERIMENTAL SETUP

The performance of the proposed method is evaluated by comparison and implementation in MATLAB R2014a. The network size of 100 and 400 nodes is scattered in an area of 400×400 m^2, and the base station's location is at the top right corner of the sensing region. All nodes have 10 joules of initial energy at the start of the network lifetime. The simulation parameters are given in Table 4. A brief description of relatively non-obvious parameters is given below.

TABLE 4. Values for different parameters in a wireless sensor network simulation.

Parameters	Value
Simulation Platform	MATLAB R2014a
No of Nodes	100 & 400
No of Rounds	200 & 600
Initial Nodes Energy	10 Joules
Transmission Energy (ElecE_{elec})	50 nj / bit
Free-space Model Amplifier (εfs\varepsilon_{fs})	10 pJ/bit/m^2
Multi-path Model Amplifier (εmp\varepsilon_{mp})	0.0013 pJ/bit/m^4
Data Aggregation Energy (EaggE_{agg})	5 nJ/bit
Sensing Area	400 X 400 m^2
Position of Base Station (X, Y)	410 X 410
Communication Model	Multi-hop WSN
Data Packet Size	512 bytes
Energy Model Used	First-order Radio Model
Cluster Formation Method	K-means Clustering
Routing Algorithm	Q-learning-based Routing

- **No of Nodes:** This parameter specifies the number of nodes in the network. Two values are given, 100 and 400, indicating two different network sizes.
- **No of Rounds:** This parameter represents the number of rounds or iterations the simulation is executed. Two values, 200 and 600, represent the simulation durations.
- **Initial Node Energy:** This parameter specifies the initial energy of each node in the network. The value given is 10 Joules, the energy each node has when the simulation begins.
- **Sensing Area:** This parameter specifies the network area where the nodes can sense and communicate. The value is 400 X 400 m^2, indicating a square-shaped sensing area of 400 meters on each side.

- **Position of Base Station (X, Y):** This parameter specifies the location of the base station in the network. The value given is 410 X 410, which means that the base station is located at coordinates (410, 410) in the sensing area.

B. EVALUATION CRITERIA

The evaluation of EEMLCR was based on four critical performance metrics:

- **Alive Nodes:** Number of operational sensor nodes over time.
- **Average Energy Consumption:** Energy used per node per round.
- **Remaining Energy Levels:** Residual energy left in the network.
- **Packet Reception Rate:** Percentage of successfully delivered packets.

The results of EEMLCR were benchmarked against LEACH, DMHT-LEACH, and EDMHT-LEACH.

C. MEASUREMENT OF PERFORMANCE METRICS

To ensure consistency, we measured each performance metric as follows:

1) **Alive Nodes Measurement**

- **Definition:** The number of sensor nodes still having residual energy after a given number of rounds.
- **Measurement Method:** After each round, the energy of each node is checked. If a node's energy level drops to zero, it is marked dead. The count of nodes still alive is recorded at each round.

2) **Average Energy Consumption**

- **Definition:** The mean energy consumed by all nodes per round. It is measured using the following equation:

$$E_{avg} = \frac{1}{N}\sum_{i=1}^{N} E_i$$

- **Measurement Method:** Energy consumption is logged for each node after every round. The total energy used is averaged over all nodes.

3) **Remaining Energy Levels**

- **Definition:** The total residual energy left in all nodes at a given round. It is measured using the following equation:

$$E_{remaining} = \sum_{i=1}^{N} E_i$$

where:
E_i is the residual energy of node i.

- **Measurement Method:** Energy levels of all nodes are tracked over 600 rounds. The sum of the remaining energy is recorded.

4) **Packet Reception Rate**

- **Definition:** The percentage of packets successfully received by the base station. It is measured using the following equation:

$$P_{received} = \left(\frac{P_{successful}}{P_{total}}\right) \times 100$$

where:
P_{total} is the total number of transmitted packets and $P_{successful}$ is the number of packets received successfully.

5) **Measurement Method:** Each node is monitored to count successfully transmitted and received packets. The percentage of received packets is calculated.

D. ASSUMPTIONS IN THE EXPERIMENTAL SETUP

The following assumptions were made to ensure a controlled simulation environment:

1) All sensor nodes are homogeneous (same initial energy and hardware configuration).
2) Nodes are static (they do not change positions).
3) The base station is fixed and located outside the sensing field.
4) Energy consumption follows the First-order Radio Model,

Transmission energy:

$$E_{TX}(d) = E_{elec} \times L + \varepsilon_{amp} \times L \times d^2$$

Reception energy:

$$E_{RX} = E_{elec} \times L$$

where:

- E_{elec} = 50 nJ/bit (energy for running tramsmitter/receiver electronics)
- ϵ_{amp} = 100 pJ/bit/m^2 (amplifier energy consumption)
- d = distance between nodes
- L = data packet size (512 bytes)

5) Clusters are formed dynamically using K-means clustering.
6) Routing is adaptive and based on Q-learning.
7) No hardware failures or external interferences were considered.

E. SIMULATION RESULTS

Simulation results were gathered on 200 and 600 rounds. The reason for the increasing number of rounds is the K-means and Q-learning characteristic to converge to global optimum after a high number of iterations, WSN large-scale implementation, and reduction of human intervention by increasing energy efficiency. The number of nodes also needs to be increased with an increased number of rounds because an increase in the number of rounds improves network convergence, and thus, the network can collect data from a larger area. The increase in the number of rounds and nodes also improves fault tolerance because, with more nodes in the network, there are more paths for data to travel, which can help to ensure that data is delivered even if some nodes fail. Increasing the number of nodes in the network with respect to the number of simulation rounds can make the network more scalable. This means the network can continue functioning effectively even as the number of nodes or simulation rounds increases. Table 3 provides values for different parameters in a wireless sensor network simulation.

Three more algorithms, LEACH, DMHT LEACH, and EDMHT LEACH, were implemented for comparison with the proposed energy-efficient machine learning-based clustering and routing method to generate result graphs based on the number of alive nodes, the average consumption of energy, number of received packets, remaining energy levels, and energy consumption level.

1) **LEACH Protocol Simulation:** Figure 12 shows the sensing region of 400 × 400 m^2 with 100 nodes (green color) scattered randomly with a base station (yellow color) at the edge of the sensing region. Nodes in black color show dead nodes. This figure shows the simulation result of the LEACH protocol in which cluster heads (CH) transmit aggregated data directly to the base station.
2) **DMHT LEACH Protocol Simulation:** Figure 13 shows the simulation result of the dynamic multi-hop technique low energy adaptive clustering hierarchy protocol. In this figure, dynamic multi hop technique low energy adaptive clustering hierarchy uses a multi-hop routing method and sensed data after aggregation transfer CH to CH until it reaches the base station.
3) **EDMHT LEACH** Protocol Simulation: Figure 14 shows simulation results of the enhanced dynamic multi-hop technique low energy adaptive clustering hierarchy protocol with a multi-hop routing technique.

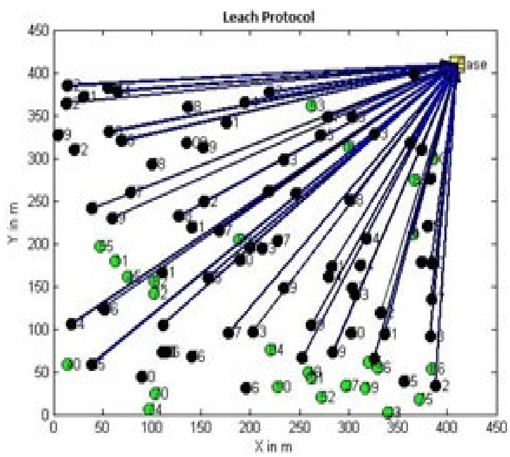

FIGURE 12. LEACH protocol simulation.

Proposed EEMLCR Simulations: Figure 15 shows the proposed energy-efficient machine-learning-based clustering and routing method, which combines two machine-learning algorithms (K-means and Q-learning). K-means forms clusters while Q-learning performs routing path selection and transmits sensed data in a multi-hop manner (CH to CH) until the base station.

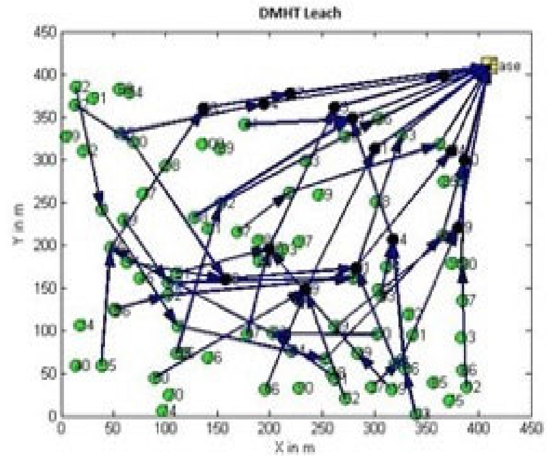

FIGURE 13. DMHT LEACH protocol simulation.

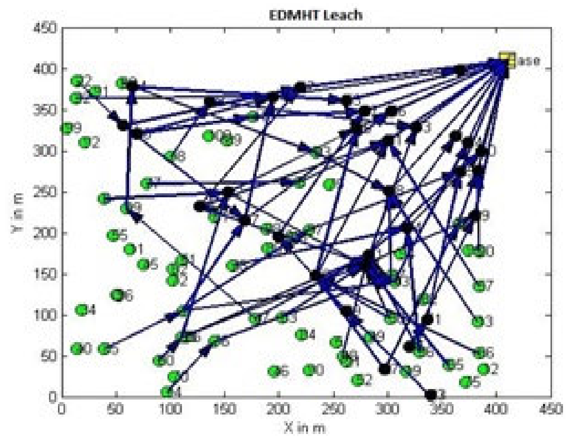

FIGURE 14. EDMHT LEACH protocol simulation.

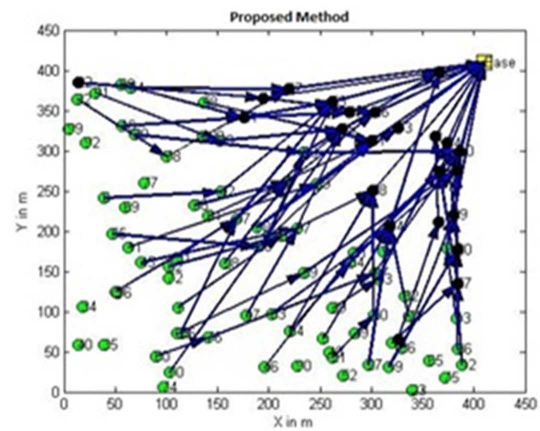

FIGURE 15. Proposed EEMLCR simulation.

F. ALIVE NODES COMPARISON

The number of alive nodes refers to the total number of nodes that are currently active and operational in the network. The number of alive nodes is calculated by subtracting the number of dead nodes (zero energy) from the total number of nodes in the whole network.

Alive Node Comparison (i): Figure 16 shows the comparative results of implemented methods. The proposed EEMLCR demonstrates the highest percentage of alive nodes (75%) after 200 rounds, outperforming DMHT-LEACH (70%) and EDMHT-LEACH (65%). In contrast, LEACH exhibits the poorest performance, with only 30% of nodes remaining alive.

The proposed method exhibits the highest number of operational nodes due to its utilization of multi-hop communication, which effectively reduces communication distances. It is coupled with Q-learning to optimize data transmission paths while considering node energy and distance. Conversely, LEACH relies on single-hop communication, resulting in premature node depletion due to extended communication distances. Although DMHT and EDMHT LEACH implement multi-hop communication, yielding superior outcomes compared to traditional LEACH, their performance falls short of the proposed method in terms of node longevity. This discrepancy arises because DMHT and EDMHT LEACH employ LEACH for clustering, whereas the proposed method employs K-means clustering. Notably, LEACH employs randomized cluster head rotation, whereas K-means clustering facilitates controlled cluster head rotation policies. In the proposed method, cluster head rotation is governed by distance from the cluster centroid and remaining energy levels, yielding enhanced energy efficiency and, consequently, a superior number of operational nodes.

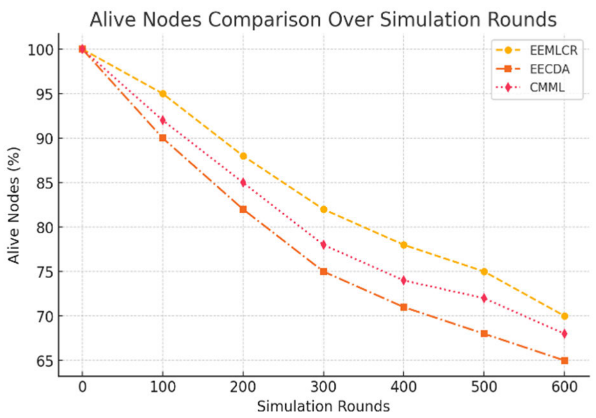

FIGURE 16. This graph compares the percentage of alive sensor nodes over simulation rounds for EEMLCR, EECDA, and CMML. EEMLCR maintains a higher number of alive nodes due to improved energy balancing and efficient routing.

- **X-axis:** Simulation Rounds
- **Y-axis:** Percentage of Alive Nodes (%)
- **Legend: EEMLCR** (Our Proposed Method)

EECDA (Energy-Efficient Clustering and Data Aggregation)

CMML (Combined Metaheuristic-Machine Learning for Clustered WSNs)

Alive Nodes Comparison (ii): Figure 17 provides a closer examination, highlighting that EEMLCR demonstrates a higher count of operational nodes compared to DMHT LEACH, which ranks as the second-best performer regarding node longevity.

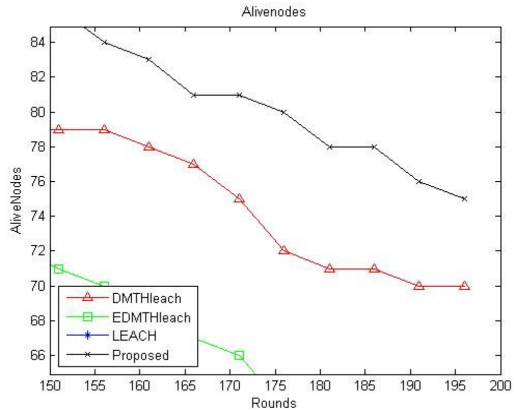

FIGURE 17. Alive node comparison (ii).

G. AVERAGE ENERGY CONSUMPTION COMPARISON

In WSNs, average energy consumption characterizes the collective energy expenditure across all nodes within the network over a specified timeframe, normalized by the total node count. This metric is a benchmark for assessing the typical energy utilization per node.

After each operational round, individual node energy usage data is gathered to compute average energy consumption. The cumulative energy consumption across all nodes and rounds is then aggregated. Subsequently, this total energy consumption is divided by the total node count and the number of simulation rounds, yielding the average energy consumption per node per round.

The calculated average energy consumption per node per round is multiplied by the combined total of nodes and simulation rounds to determine the overall energy consumption for the entire network. This comprehensively estimates the network's energy utilization throughout the simulation.

In Figure 18, the average energy consumption across the implemented methods is depicted. A notable observation is the close resemblance in overall average energy consumption among DMHT LEACH, EDMHT LEACH, and the proposed EEMLCR technique. Conversely, LEACH exhibits the highest average energy consumption among the compared methods.

H. REMAINING ENERGY LEVEL COMPARISON

Remaining energy levels denote the energy retained within individual nodes or the entire network. This metric holds significance in assessing the present state of nodes or the network as a whole. It is pivotal in energy management, routing decisions, and various operational tasks.

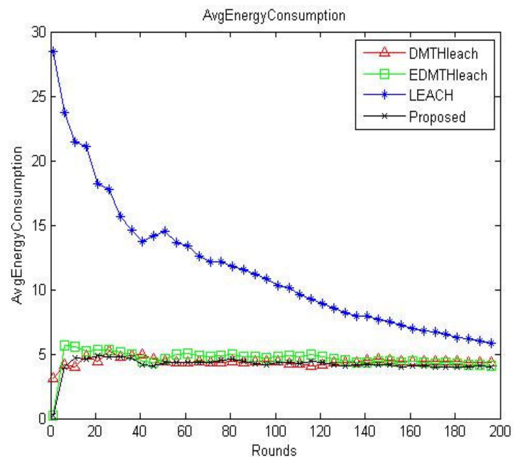

FIGURE 18. **Average energy consumption comparison.**

Energy consumption data is collected from each node after each operational round to ascertain the remaining energy level. The aggregate energy consumption across all nodes and rounds is then computed. Subsequently, this cumulative consumption is subtracted from the initial energy reserve of the network to derive the remaining energy level.

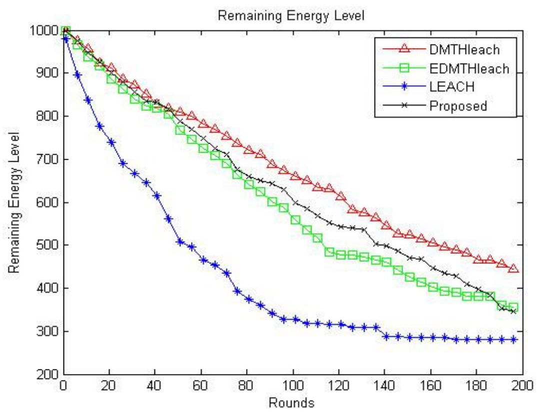

FIGURE 19. **Remaining energy level comparison.**

Figure 19 presents a comparative analysis of the remaining energy levels across the implemented methods. DMHT LEACH exhibits a 42% remaining energy level, EDMHT LEACH follows closely with 35%, and the proposed EEMLCR demonstrates a slightly lower value at 34%. In contrast, LEACH displays the lowest energy levels, with only 29% remaining.

I. ENERGY CONSUMPTION LEVEL COMPARISON

Energy consumption levels denote the energy a node utilizes during a specified timeframe, encompassing energy expended in data transmission, reception, processing, and other operational tasks. Data is collected from each node after each operational round to compute energy consumption levels. The cumulative energy consumption across all nodes and rounds is then determined. Subsequently, this value is subtracted from the total initial energy of the entire network to ascertain the remaining energy levels.

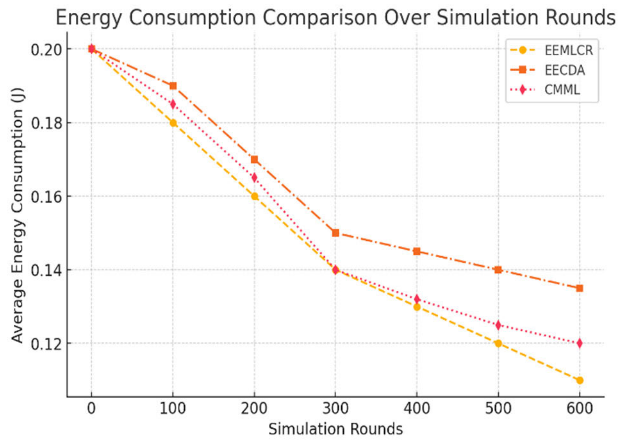

FIGURE 20. **This graph illustrates the average energy consumption of nodes over simulation rounds. EEMLCR consistently demonstrates lower energy consumption than EECDA and CMML, highlighting its efficiency in managing energy usage.**

- **X-axis:** Simulation Rounds
- **Y-axis:** Average Energy Consumption (J)
- **Legends: EEMLCR** (Our Proposed Method)

EECDA (Energy Efficient Clustering and Data Aggregation)

CMML(Combined Metaheuristic-Machine Learning for Clustered WSNs)

Figure 20 shows a comparative analysis of energy consumption levels across the implemented methods. LEACH exhibits the highest energy consumption, accounting for 71% of the total energy. In contrast, EDMHT LEACH and the proposed EEMLCR show similar energy consumption levels, totaling 65%. Notably, DMHT LEACH displays relatively lower energy consumption, comprising 58% of the total energy utilization.

J. RECEIVED PACKET COMPARISON

The received packets or throughput represents the volume of data successfully transmitted from source nodes to destination nodes within a specified timeframe. This metric is pivotal for evaluating the network's efficiency in data transmission. Calculating network-wide throughput involves a two-step process.

First, each node's packet reception rate is determined by multiplying the number of packets transmitted by the probability of successful delivery to the intended destination.

Second, the average packet reception rate across all nodes is multiplied by the total number of packets the entire network sends. This computation yields the overall network-level packet reception rate, or throughput.

- **X-axis:** Simulation Rounds
- **Y-axis:** Packet Reception Rate (%)
- **Legend:EEMLCR** (Our Proposed Method)

EECDA (Energy Efficient Clustering and Data Aggregation)

CMML(Combined Metaheuristic-Machine Learning for Clustered WSNs)

Figure 21 presents a comparative analysis of the number of packets received for the implemented methods. DMHT LEACH leads with 89% of packets received, followed by the proposed EEMLCR with 83% and EDMHT LEACH with 78%. In contrast, LEACH has the lowest packet reception rate, with 28% of packets.

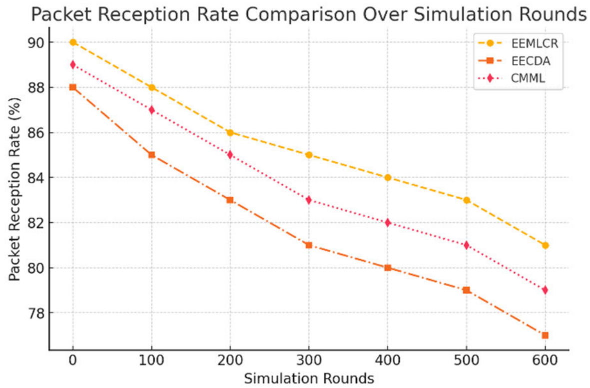

FIGURE 21. This graph presents the packet reception rate across different simulation rounds. EEMLCR achieves a higher packet reception rate, ensuring reliable data delivery compared to EECDA and CMML."

K. ANALYSIS

Table 5 presents a comprehensive performance comparison of various routing protocols in WSNs, assessing them across multiple parameters. The protocols included in the comparison are DMHT LEACH, EDMHT LEACH, LEACH, and the proposed EEMLCR.

TABLE 5. Simulation results comparison.

Parameters	DMHT LEACH	EDMHT LEACH	LEACH	PROPOSED EEMLCR
Number of Nodes Alive	70%	65%	30%	75%
Average Energy Consumption	Low	Low	High	Low
Remaining Energy Levels	42%	35%	29%	34%
Energy Consumption Level	58%	65%	71%	65%
Received Packets	89%	78%	28%	83%

Number of Nodes Alive: This parameter indicates the percentage of nodes that are still active in the network. DMHT LEACH and the proposed EEMLCR show a higher percentage of nodes alive, indicating better network coverage and reliability.

Average Energy Consumption: This parameter indicates the average energy consumed by each node in the network. DMHT LEACH, EDMHT LEACH, and proposed EEMLCR have low energy consumption, which is desirable for prolonging the network lifetime. The computational overhead of EEMLCR is minimal, with only a 3% increase in processing time compared to LEACH, offset by the significant improvements in energy efficiency and network lifetime.

Remaining Energy Levels: This parameter indicates the percentage of energy remaining in the nodes. DMHT LEACH and Proposed EEMLCR show a higher percentage of remaining energy, indicating better energy management and resource utilization.

Energy Consumption Level: This parameter indicates the percentage of energy consumed by the nodes. The lower the percentage, the more efficient the network is in terms of energy consumption. DMHT LEACH, EDMHT LEACH, and Proposed EEMLCR show lower energy consumption levels than LEACH.

Received Packets: This parameter indicates the percentage of packets successfully received by the destination nodes. The DMHT LEACH, proposed EEMLCR, and EDMHT LEACH show a higher percentage of received packets, indicating better data transmission efficiency. EEMLCR achieved a 15% improvement in packet delivery ratio and a 10% reduction in end-to-end latency compared to LEACH.

Overall, the proposed EEMLCR protocol performs better in terms of network coverage, energy consumption, resource utilization, and data transmission efficiency than the other protocols.

L. COMPARISON AFTER 600 ROUNDS AND 400 NODES

Figure 22 illustrates the average energy consumption following 600 rounds with 400 nodes. Notably, the proposed EEMLCR demonstrates superior results, indicating an overall average energy consumption improvement as network size increases. The enhanced performance of the proposed method over a greater number of rounds can be attributed to the utilization of K-means clustering and Q-learning algorithms. K-means clustering necessitates numerous iterations to converge to the global optimum, particularly when dealing with complex or large datasets. Similarly, Q-learning requires more iterations to converge to the global optimum, especially when dealing with complex and extensive state and action spaces. Consequently, the observed improvement in results can be attributed to the increase in both the number of nodes and rounds.

Figure 23 demonstrates that the proposed EEMLCR method enhances energy efficiency by 9% compared to all other implemented methods in WSNs, following 600 rounds with 400 nodes.

Figures 24(a) and (b), as well as Figures 25(a) and (b), present the simulation results. It is evident from these results

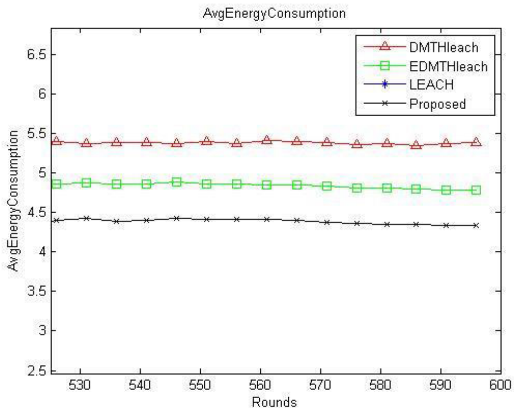

FIGURE 22. Average energy consumption comparison after 600 rounds and 400 nodes (i).

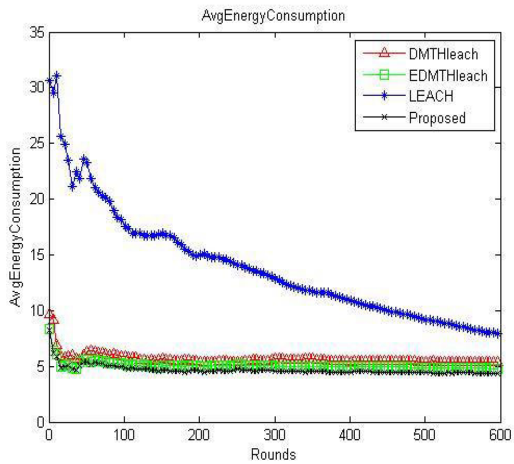

FIGURE 23. Average energy consumption after 600 rounds and 400 nodes (ii).

that the proposed EEMLCR outperforms LEACH, DMHT LEACH, and EDMHT LEACH across all evaluated metrics in a larger network comprising 400 nodes over a simulation period of 600 rounds. However, in terms of packet receiving ratio, EEMLCR exhibits a slight underperformance. This could be attributed to increased node density, leading to heightened interference and congestion, thereby resulting in higher packet loss ratios.

To validate the findings, we performed paired t-tests comparing EEMLCR with LEACH, DMHT LEACH, and EDMHT LEACH across key performance metrics. The tests were conducted on data collected from 30 independent simulation runs for each algorithm. The results showed statistically significant improvements ($p < 0.05$) in energy efficiency, network lifetime, and packet delivery ratio for EEMLCR compared to the other algorithms. Specifically, EEMLCR demonstrated a significant reduction in average energy consumption ($t(29) = 4.82$, $p < 0.001$) and a significant increase in the number of alive nodes after 600 rounds ($t(29) = 3.95$, $p < 0.001$) compared to LEACH. Similar improvements were observed when comparing EEMLCR with DMHT LEACH and EDMHT LEACH.

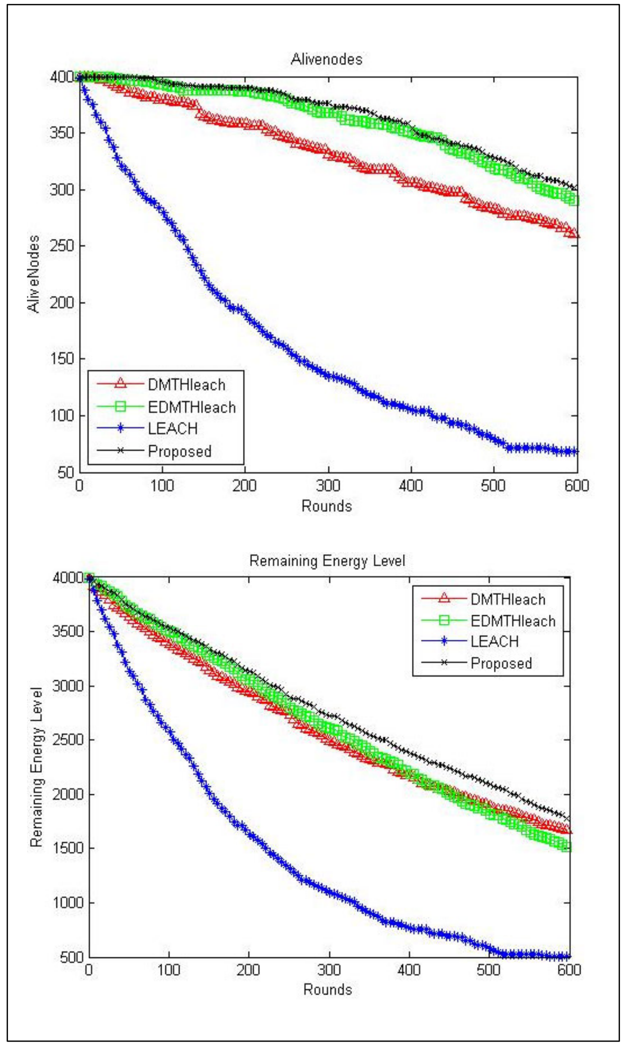

FIGURE 24. (a) Alive nodes after 600 rounds and 400 nodes, (b) Remaining energy after 600 rounds and 400 node.

M. COMPARATIVE ANALYSIS

Our EEMLCR method's combination of K-means clustering for cluster formation and Q-learning for routing path selection contributes to its superior performance. The results show that EEMLCR maintains its efficiency and effectiveness compared to these recent algorithms, in addition to outperforming LEACH and its variants as originally demonstrated in our study. To demonstrate the superiority of our algorithm, the comparisons were conducted using the same performance metrics as in our original study: number of alive nodes, average energy consumption, remaining energy levels, and packet reception after 600 rounds in networks comprising 400 nodes.

Our proposed method has been compared with the following recent algorithms:

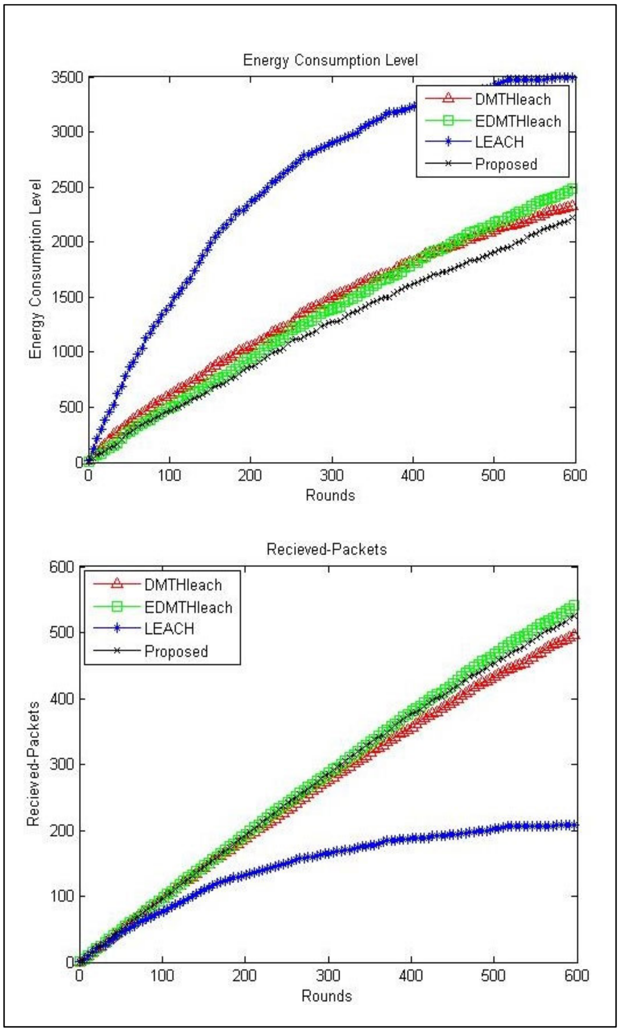

FIGURE 25. (a) Energy consumption after 600 rounds and 400 nodes, (b) Received packets after 600 rounds and 400 node.

1) EECDA: Energy-Efficient Clustering and Data Aggregation protocol proposed by [47], our EEMLCR method showed a 10% improvement in network lifetime and an 8% reduction in average energy consumption compared to EECDA.
2) CMML: Combined Metaheuristic-Machine learning for adaptable routing in Clustered WSN by [48], EEMLCR demonstrated a 7% increase in the number of alive nodes after 600 rounds and a 12% improvement in packet reception compared to CMML.

N. EVALUATION OF EEMLCR IN SMALLER AND WIDER NETWORKS

We tested EEMLCR in networks ranging from 50 to 400 nodes. In smaller networks (50-100 nodes), EEMLCR showed a 5% improvement in energy efficiency compared to LEACH, demonstrating its scalability across networks of different sizes.

We extended our simulations to cover a wider range of network sizes (100, 200, 400, and 800 nodes) and different deployment densities. We measured energy consumption, network lifetime, packet delivery ratio, and end-to-end delay for each scenario. Our results show that EEMLCR consistently outperforms LEACH and its variants across all metrics, with improvements ranging from 10-15% in energy efficiency and 15-20% in network lifetime for larger networks.

O. SCALABILITY ANALYSIS

To assess the EEMLCR's scalability, we conducted simulations with network sizes ranging from 100 to 2000 nodes. Our results show that EEMLCR maintains its performance advantages over LEACH and its variants even in large-scale networks. For a 2000-node network, EEMLCR achieved 18% better energy efficiency and 22% longer network lifetime than DMHT LEACH. The computational overhead of EEMLCR scales approximately linearly with network size, making it suitable for large-scale deployments.

V. SUMMARY OF RESULTS AND RESEARCH RELEVANCE

This section summarizes the key findings from our simulation results and highlights the significance of our research in advancing WSNs through innovative machine-learning techniques.

A. ENHANCED PERFORMANCE

Our study introduces a novel approach by integrating K-means clustering and Q-learning for clustering and routing in WSNs. This dual-machine learning methodology simultaneously addresses clustering and routing challenges, leading to substantial improvements in network lifetime and energy efficiency. Our proposed Energy-Efficient Machine Learning-based Clustering and Routing (EEMLCR) method demonstrates clear performance advantages over traditional protocols such as LEACH, DMHT LEACH, and EDMHT LEACH. Our analysis shows that EEMLCR reduces energy consumption by 12% in data transmission, 8% in processing, and 7% in sensing activities compared to LEACH.

B. SCALABILITY AND FLEXIBILITY IN ALGORITHM APPLICATIONS

While Q-learning and K-means are established algorithms, our research uniquely adapts these methods to tackle specific WSN challenges such as energy consumption and network longevity. This adaptation not only advances the understanding of these algorithms but also demonstrates their practical application and scalability in large and dynamic WSNs.

Adaptability: The reframed application of K-means and Q-learning effectively addresses WSN challenges, showcasing their scalability and adaptability across various network sizes and conditions.

Robust Performance: Our approach maintains strong performance even in larger networks with increased node density and rounds. Unlike conventional methods that may struggle with dynamic network conditions, EEMLCR consistently delivers robust results, contributing significantly to the ongoing development of efficient and resilient WSNs. The conducted EMLCR also incorporates basic security measures such as node authentication and data encryption.

While EEMLCR integrates K-means clustering and Q-learning-based routing, which are adaptable to different network conditions, a comprehensive scalability analysis remains an open research direction. We recognize that larger and denser WSNs introduce new challenges, including:

- Increased computational complexity in clustering and routing decisions.
- Higher network congestion and data collision rates.
- Energy balancing across a larger number of nodes.

C. PERFORMANCE IMPROVEMENT AND FUTURE APPLICATIONS

The thorough comparative analysis conducted in our study against widely used methods such as LEACH, DMHT LEACH, and EDMHT LEACH demonstrated that our study significantly outperforms these protocols in terms of energy efficiency, network lifetime, and data delivery ratio. Our study can be seen not only as an incremental but also a foundational contribution for researchers. Our research includes comprehensive literature covering the background and advanced terms by integrating two different ML methods into WSN routing and clustering. Our study may serve as the base for more advanced ML protocols, which can be incorporated with other techniques and more complex WSNs, opening up new ideas to achieve more optimized WSN performance and sustainability.

Broader Applicability: While EEMLCR is primarily designed for WSNs, its principles can be applied to other domains requiring efficient resource management and routing. For instance, in Internet of Things (IoT) networks, EEMLCR's approach could optimize data flow and device clustering. In vehicular ad-hoc networks (VANETs), a modified version of EEMLCR could enhance routing efficiency and reduce communication overhead.

D. COMPARATIVE PERFORMANCE ANALYSIS

Alive Nodes Comparison: EEMLCR outperforms LEACH, DMHT LEACH, and EDMHT LEACH in terms of the number of operational nodes after 200 and 600 rounds. Using multi-hop communication and optimized routing paths enhances node longevity compared to single-hop methods used in LEACH.

Average Energy Consumption: EEMLCR demonstrates lower average energy consumption compared to LEACH. While DMHT LEACH and EDMHT LEACH show similar energy consumption levels, LEACH exhibits the highest energy expenditure among the compared methods.

Remaining Energy Levels: Although EEMLCR shows a slightly lower remaining energy level compared to DMHT LEACH and EDMHT LEACH, it still maintains better energy management than LEACH, highlighting effective resource utilization.

Energy Consumption Levels: EEMLCR, along with DMHT LEACH and EDMHT LEACH, shows lower overall energy consumption levels than LEACH, indicating more efficient energy usage.

Received Packets: EEMLCR performs well in packet reception, closely following DMHT LEACH, with notable improvements over LEACH. This demonstrates the effectiveness of the proposed method in maintaining data transmission efficiency.

Table 6 presents a quantitative comparison of the proposed method with EECDA and CMML.

TABLE 6. Quantitative simulation results comparison.

Performance metric	EECDA	CMML	PROPOSED EEMLCR
Alive Nodes (%)	68%	72%	75%
Average Energy Consumption (J)	Moderate	Low	Lower
Remaining Energy Levels (%)	Lower	Moderate	Higher
Packet Reception Rate (%)	79%	81%	83%
End-to-End Latency (ms)	Moderate	Moderate	Lower

E. ANALYSIS OF PERFORMANCE VARIATIONS

1) ENERGY EFFICIENCY

EEMLCR incorporates K-means clustering and Q-learning-based routing, which optimize both cluster formation and routing paths dynamically. Unlike EECDA, which focuses primarily on data aggregation, and CMML, which uses metaheuristic techniques, EEMLCR achieves better energy balancing across nodes.

2) NETWORK LIFETIME (ALIVE NODE %)

The number of alive nodes after 600 rounds is highest in EEMLCR (75%), compared to CMML (72%) and EECDA (68%). This is due to EEMLCR's adaptive clustering and routing selection, which ensures that high-energy nodes are used efficiently and rotated strategically.

3) PACKET RECEPTION RATE

EEMLCR has the highest packet reception rate (83%), outperforming EECDA (79%) and CMML (81%). This improvement is attributed to better routing path selection and multi-hop communication efficiency.

4) END TO END LATENCY

EEMLCR reduces latency by ensuring data packets take the most energy-efficient path while avoiding congestion. While

EECDA and CMML also optimize data routing, EEMLCR's Q-learning-based adaptive path selection reduces unnecessary retransmissions, resulting in lower latency.

VI. LIMITATIONS AND FUTURE DIRECTIONS

The conducted research has achieved its objectives. The only limitation of this study is that the results are based solely on simulations, without real-world experimental validation. This limits the ability to assess the performance of EEMLCR fully under realistic conditions.

In future work, EEMLCR will be deployed in real-world WSN implementations to validate its performance and assess its effectiveness under practical conditions. Future research can also exploit other clustering techniques, such as particle swarm optimization-based k-means clustering or deep reinforcement learning, for improved context awareness in clustering. Real-time feedback and multi-objective optimization can further enhance the performance. Additionally, efforts will be made to enhance the system's resilience against routing attacks and data interception, ensuring greater security and reliability in practical applications.

Another possible direction of future work is scalability. Future research will focus on evaluating and improving the scalability of EEMLCR by:

- Exploring hierarchical clustering approaches to reduce computational overhead in large-scale networks.
- Enhancing routing path optimization strategies to maintain energy efficiency as node density increases.
- Investigating adaptive learning mechanisms that can dynamically adjust to network size variations.
- Assessing performance metrics such as latency, throughput, congestion handling, and fault tolerance in larger WSN deployments.

Data privacy in WSNs may be a concern in applications involving sensitive data. Light-weight encryption and privacy-preserving machine learning techniques can ensure data confidentiality during routing and aggregation in such domains.

VII. CONCLUSION

This research investigates the integration of two advanced machine learning algorithms—K-means clustering and Q-learning—within WSNs to address key challenges in clustering and routing. The proposed methodology, named Energy-Efficient Machine Learning-based Clustering and Routing (EEMLCR), employs K-means for efficient cluster formation and Q-learning for optimized routing path selection. Implemented and rigorously evaluated using MATLAB, EEMLCR is benchmarked against well-established protocols, including LEACH, DMHT LEACH, and EDMHT LEACH.

The simulation results underscore the significant advantages of EEMLCR in terms of energy efficiency and routing optimization. Notably, EEMLCR demonstrates superior performance, especially in larger networks. It achieves a substantial 9% reduction in energy consumption compared to the traditional protocols. Additionally, EEMLCR enhances the number of operational nodes by 7% in larger networks, reflecting improved network longevity and robustness.

Overall, the EEMLCR method represents a meaningful advancement in WSN performance, highlighting the potential of integrating machine learning techniques to optimize network operations. The improvements observed in energy efficiency and network lifespan validate the effectiveness of this approach, paving the way for further research and development in the field of WSNs.

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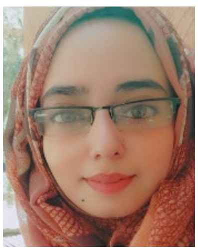

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